

CHAPTER 20

PROGRAM DESIGN AND TECHNIQUE FOR SPEED AND AGILITY TRAINING

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After completing this chapter, you will be able to

- describe the underlying biomechanical constructs of sprint, change-of-direction, maneuverability, and agility performance;
- apply fundamental biomechanical movement principles to the coaching of locomotion modes and techniques;
- analyze the abilities and skills needed to perform specific movement tasks;
- effectively monitor the development of sprint, change-of-direction, and agility abilities;
- apply evidence-based methods for developing speed, change of direction, and agility; and
- design and implement training programs to maximize athletic performance.

This chapter addresses the development of speed, change-of-direction, and agility abilities. Although the term **speed** is often used when an athlete displays one or all of these aspects of physical performance, it is critical to understand that athlete development requires various underpinning physical capacities and skills as a result of differing biomechanical requirements. These three important aspects of physical performance can be defined as follows:

- *Speed*. The skills and abilities needed to achieve high movement velocities
- *Change of direction*. The skills and abilities needed to explosively change movement direction, velocities, or modes
- *Agility*. The skills and abilities needed to change direction, velocity, or mode in response to a stimulus

The ability to outrun and maneuver the competition is a hallmark of most athletic endeavors. Further, the ability to change direction rapidly during an activity can mute the effect of an opponent's speed or provide both physical and tactical advantages on the field of play. Although it may seem that these scenarios all involve the "speed" of an athlete, this perceived speed may be a result of one quality or a combination of the three qualities just listed. Within sport, high-speed human locomotion can be categorized as linear or multidirectional. Generating high speeds linearly, more commonly referred to as *sprinting*, is the underlying requirement for success in many track and field events as well as open-field running in game-based contexts. While linear speed is important in team sports, play is primarily multidirectional. As a result, these athletes' success partially depends on responding to ever-changing game scenarios through fast, efficient changes of direction and decelerations, whereas speed requires the ability to accelerate and reach maximal velocity.

Because of the nature of most sports, there are scenarios in which athletes have a predetermined change of direction to make

and they are limited only by their physical capacity to perform this activity (e.g., a route, play, or predetermined pattern). The physical capacity to change direction while decelerating and then reaccelerating, sometimes using a different mode of travel, is **change-of-direction (COD)** ability, whereas **agility** requires the use of perceptual-cognitive ability in combination with change-of-direction ability. This highlights the similar and contrasting qualities of speed, change of direction, and agility. For example, **acceleration** is a part of change-of-direction ability and agility, but additional aspects such as **deceleration** ability and mode of movement differentiate the training for speed from the training for change of direction and agility. The physical capacity to change direction may be a component of agility (i.e., action capacity), but the perceptual-cognitive component influences the physical demands of agility. Therefore, while reading this chapter, one should understand that although these qualities overlap some, differing physical, technical, or perceptual-cognitive development is required to improve each of these aspects of physical performance.

When athletes are sprinting or changing direction, their performance is a function of physical capacity and technical proficiency. While biomechanical and metabolic efficiency underpin performance in aerobic sport, effective application of force (net impulse) limits speed, change of direction, and agility. Simply put, an athlete's success in these explosive movements is the product of an athlete's strength capacity combined with the ability to use this strength within the constraints of the activity. **Strength** is often associated with an athlete's capacity to produce force, but it is important to understand that while high levels of maximal strength are desirable attributes in sport, sprint, COD, and agility events occur in periods that prevent athletes from producing and expressing their maximal strength, thus the rate at which net forces are produced is essential.

Within sprinting, the application of force allows the athlete to accelerate, attain high velocities, and attempt to maintain the high

velocities. In addition to the force application associated with acceleration and attainment of velocity in sprinting, COD ability requires effective application of force to decelerate, deflect the center of mass, and then reaccelerate in another direction. Further still, agility performance is considered a function of the athlete's ability not only to change direction but also specifically to change direction in response to a stimulus (4) such as a defender or ball; this is known as **perception–action coupling**. For this reason, strength and conditioning professionals must be aware of their training choices regarding the development of physical characteristics that assist in promoting speed, COD, and agility on both the track and the field.

Speed requires the ability to accelerate and reach maximal velocity, whereas agility performance requires the use of perceptual- cognitive ability in combination with the ability to decelerate and then reaccelerate in an intended direction.

Speed and Agility Mechanics

To execute movement techniques, athletes must apply force—the product of mass and acceleration. Due to the limited available time to produce force during most athletic activities, two variables describe force relative to the time available to produce force:

- **Rate of force development (RFD)**—The development of maximal force in minimal time, typically used as an indicator of explosive strength (3).
- **Impulse**—The product of the generated force and the time required for its production, which is measured as the area under the force–time curve. According to the **impulse–momentum relationship**, net impulse dictates the magnitude of change of momentum of an object. Because mass remains constant in most sporting scenarios, increases in net impulse are therefore proportionate to changes in velocity.

Physics of Sprinting, Change of Direction, and Agility

Force represents the interaction of two physical objects. Force is a vector quantity, meaning that it has both magnitude (size) and direction. Traditionally, force is described as a push or a pull exerted on one object by another, which prevents both objects from occupying the same space. This movement of mass changes an object's velocity, causing acceleration.

Within the strength and conditioning profession, **velocity** and speed are often used interchangeably. For a proper discussion on sprint and agility performance, these terms need to be separated. Speed is a **scalar quantity**, which means that it describes only how fast an object is moving. Speed is the rate at which an object covers a distance. Like force, velocity is a **vector quantity**. Velocity describes both how fast an object is traveling and its direction. In short, velocity is speed with a direction.

Acceleration refers to the rate at which an object's velocity changes over time. Once a force acts on a physical object, the mass will change direction and leave the space it was occupying. Acceleration of the object will continue if external forces continue to change velocity. In practical settings, deceleration replaces **negative acceleration** in describing a change from higher to lower velocity.

Rate of Force Development

Within the sporting context, the ability to produce force rapidly is arguably a more desirable trait than maximal force production given the time constraints associated with most high-intensity actions (100). While the ability to generate high levels of maximal force has been shown to improve performance in jump height and other athletic measures, most competitive scenarios do not occur within a time frame that allows an athlete to generate maximal forces (27). Specifically, the generation of maximal contraction force takes at least 300 milliseconds, while many sport activities consume 0 to 200 milliseconds ([figure 20.1](#)) (1). For this reason, in sport settings where success is restricted to timing of movement, RFD is a more useful measure of an athlete's impulsive ability (5). RFD can be described as the change in force divided by the change in time (100).

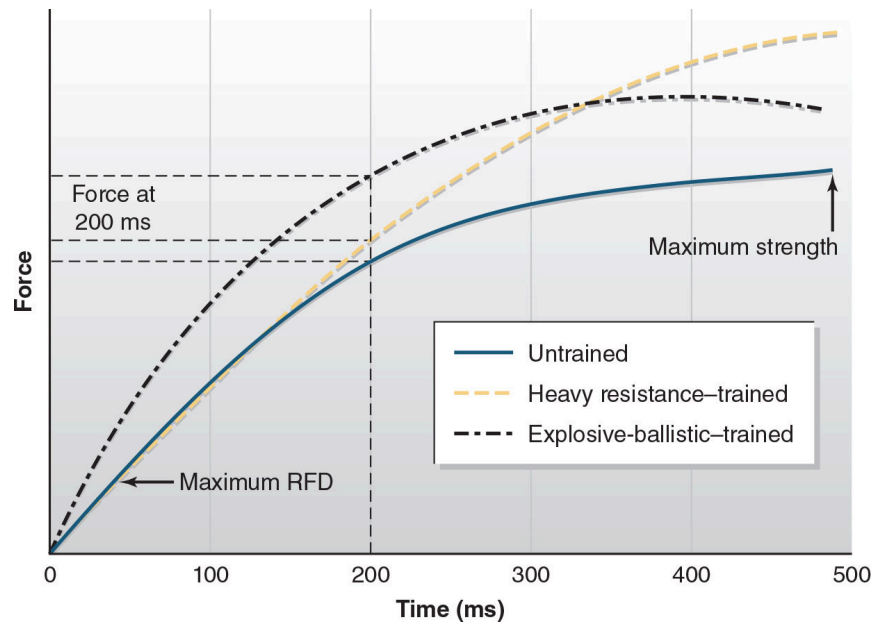


FIGURE 20.1 Force as a function of time, indicating maximum strength, rate of force development (RFD), and force at 0.2 seconds for untrained (solid blue line), heavy resistance-trained (dashed yellow line), and explosive-ballistic-trained (dotted black line) subjects. Impulse is the change in momentum resulting from a force, measured as the product of force and time (represented by the area under each curve), and is increased by improving RFD. When functional movements are performed, force is typically applied very briefly—often for 0.1 to 0.2 seconds—whereas absolute maximum force development may require 0.6 to 0.8 seconds.

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The ability to accelerate a mass depends on a change in velocity resulting from the application of an external force. Therefore, from a practical standpoint, an athlete wishing to achieve higher acceleration capabilities should apply forces at a greater rate and orient force in the optimal direction (84).

Impulse

For an object to change its position, forces must be applied (to overcome inertia) to produce a change in velocity. Athletes attempting to increase locomotor speeds through the production of force never apply force instantaneously. In fact, force is applied to

the running surface over a period of time (i.e., impulse) in the stance phase of sprinting ([figure 20.2](#)) or in the plant phase of changing direction. The length of time athletes are in this stance or plant phase is termed their **ground contact time**. The product of the time the force is applied to the ground and the amount of force applied is termed impulse and can be graphically represented as the area under a force–time curve. Changes in impulse result in changes of the athlete’s momentum and therefore the ability to either accelerate or decelerate, thus underpinning human locomotion.

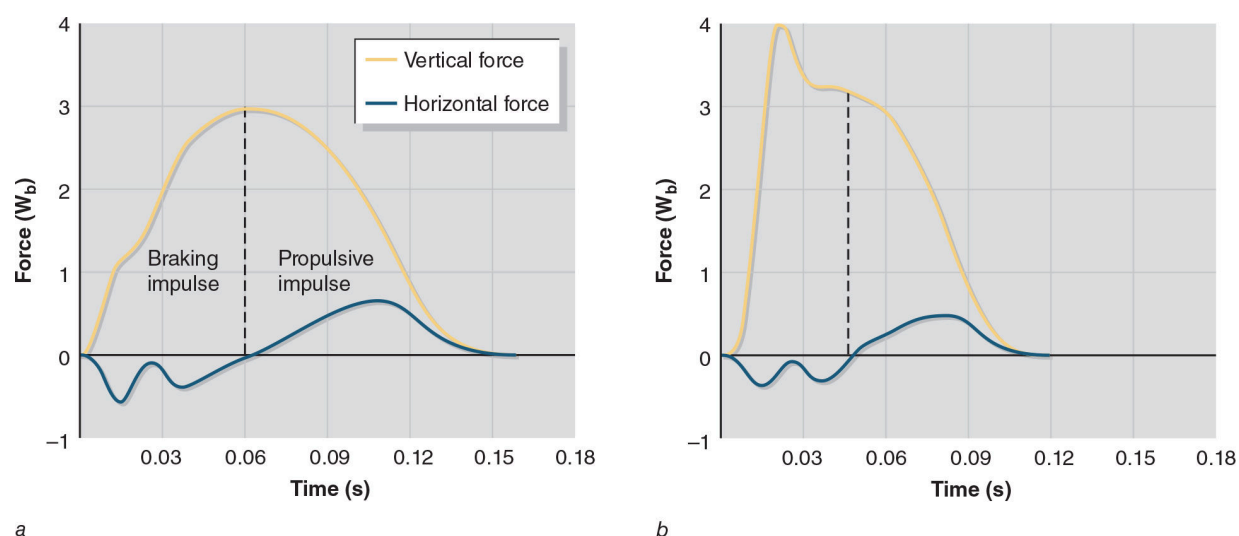


FIGURE 20.2 Sprint ground reaction force and impulse during the (a) acceleration phase and (b) maximal-velocity phase. W_b = body weight.

[Figure 20.2](#) demonstrates how the vertical and horizontal forces differ in magnitude when the acceleration phase ([figure 20.7](#) on [page 645](#)) is compared to the maximal-velocity phase ([figure 20.8](#) on [page 646](#)), which reflects the two different body positions used to produce force through maximal extension. Further, these two diagrams demonstrate the concept of impulse (represented as the area under the force–time curve) during the **braking phase**, indicated by a negative horizontal force, or during the **propulsive phase** that occurs where there is a positive horizontal force. These two phases of impulse are indicated by the vertical dotted line. It

should be noted that during maximal-velocity phases, production of force is asymmetrical and the RFD is very high, resulting in much shorter ground contact times in comparison to the acceleration phase.

Momentum is defined as the product between the mass of an object and the velocity of movement. During a sprint, an athlete's body mass remains constant. Thus, given the same time frame, the only way to achieve a greater impulse is to generate greater net force. This increase in net impulse leads to an increase in momentum or a decrease in momentum, depending on whether the athlete is intending to accelerate or reaccelerate or decelerate before a COD or as an isolated action. In other words, a change in impulse results in a change in momentum and is the cause of an object's movement.

Within human locomotion, the magnitude of the force coupled with the length of time the force is produced during an individual step is paramount to success. Changes in these forces can increase or decrease the athlete's momentum. For this reason, training should focus on developing impulse—the area under the force–time curve—in addition to RFD.

Power has not been discussed here because it is derived from force and velocity. Therefore, power is considered a mechanical construct and, practically speaking, it can be more useful to directly measure the characteristics that determine power production (e.g., force and velocity), because measurement of power alone does not give insight into the performance in a way that is fully useful to practitioners. In understanding that force, RFD, and impulse are direct, underlying characteristics that dictate explosive performance, one does not require a more complex derived value to gain additional insight.

Practical Implications for Speed

To displace their body mass down a track or field, athletes must produce forces sufficient to overcome the effects of gravity and

create a positive change in velocity. Within a short sprint, force is the effort needed to accelerate an athlete up to his or her highest achievable speeds, which are largely determined by physiological factors. These forces or efforts are produced rapidly, with time constraints that are shorter than the time needed for maximal voluntary force production. For this reason, rate of force production may be a more important factor for sprinting success. Moreover, since sprinting success is largely dependent on the production of forces within a short amount of time, impulse is an important underlying factor.

Practical Implications for Change of Direction and Agility

In addition to the requirement for acceleration, the production of braking forces over certain periods of time, or foot contacts, termed **braking impulse**, should be considered during COD, deceleration, and agility maneuvers. The amount of impulse required to change momentum effectively and efficiently is a direct reflection of the physical requirements for COD. For example, as the angle of directional change required typically increases or the entry velocity into the COD increases, so does the impulse required to change momentum and deflect the center of mass; therefore, it is physically more demanding to perform such activities. Further, the time restraints placed on a performer due to the perceptual-cognitive aspects of agility can influence the physical demands by limiting the time available to produce the required force (and impulse) to successfully change direction in response to a stimulus.

Speed requires the ability to accelerate and reach maximal velocity, whereas agility performance requires the use of perceptual-cognitive ability in combination with the ability to decelerate, change direction, and then reaccelerate in an intended direction.

Neurophysiological Basis for Speed

Sprinting, agility, and COD are all dynamic displays of force production in the athletic setting. Because strength and conditioning professionals are often asked to assist in the development of these competitively advantageous qualities, an overview of how these force measures are produced during movement is warranted.

Nervous System

Neuromuscular function is vital to sprint performance because the activity and the interaction of the central nervous system with the muscles ultimately influence the rate and strength of muscle contraction. Research has shown that the combination of strength, plyometric, and sprint training produces several adaptations within the neuromuscular system that may contribute to improved sprint performance. Resistance training enhances neural drive, the rate and amplitude of impulses being sent from the nervous system to the target muscles (1). Increases in neural drive, which are indicative of an increase in the rate at which action potentials occur, are related to increases in both muscular force production and the rate of force production. Similarly, plyometric training demonstrates increases in excitability of high-threshold motor neurons. Increased excitability ultimately enhances neural drive to the muscle. Taken together, increases in neural drive may contribute to increases in the athlete's RFD and impulse generation.

Stretch-Shortening Cycle

Many functional tasks begin with preparatory countermovements involving spring-like actions referred to as the **stretch-shortening cycle (SSC)**—an eccentric–concentric coupling phenomenon in which muscle–tendon complexes are rapidly and forcibly lengthened, or stretch loaded, and immediately shortened in a reactive or elastic manner, resulting in enhanced concentric force production. Practically speaking, the SSC is demonstrated in movements in which

a rapid transition from an eccentric action to a concentric action occurs. Therefore, SSC actions are particularly prevalent in sports involving running, jumping, directional changes, and other explosive changes in velocity; thus reactive strength development is a highly important quality to develop in training programs and can be a distinct capability that is independent of maximal strength in elite athletes (45, 55, 56, 82, 83, 91, 98, 112).

SSC actions exploit intrinsic muscle–tendon behavior as well as the force and length reflex feedback to the nervous system (3, 4, 10, 18, 33). Acutely, SSC actions tend to increase mechanical efficiency and impulse via elastic energy recovery, whereas chronically, they upregulate muscle stiffness and enhance neuromuscular activation (45, 55, 56, 82, 83).

Training activities aimed at improving SSC performance should fulfill two criteria (47, 82, 83, 91, 98):

- They should involve skillful, multijoint movements that transmit forces through the kinetic chain and exploit elastic–reflexive mechanisms.
- To manage fatigue and emphasize work quality and technique, they should be structured around brief work bouts or clusters separated by frequent rest pauses.

In practice, a combination of progressive plyometric and heavy resistance methods can accomplish these objectives (i.e., complex or contrast training).

Spring–Mass Model

Exposure to strength and speed training may be linked to a rise in the preactivation of the musculature used in sprinting (54, 56). The onset of pre-tension may be related to an increase in the sensitivity of associated muscle spindles. The improvement in the time needed for feedback from muscle spindles results in greater muscle stiffness and tendon compliance (54, 58, 79). This physiological condition provides support to the SSC, which underpins the **spring–mass**

model (SMM). The SMM ([figure 20.3](#)) is a mathematical model that depicts sprinting as a type of human locomotion in which the displacement of a body mass is the aftereffect from energy produced and is delivered through the collective coiling and extension of spring-like actions within muscle architecture (11, 29, 37, 39). During a complete running cycle, one spring compresses and propels the sprinter's body forward. Simultaneously, the other spring swings forward in preparation for ground contact.

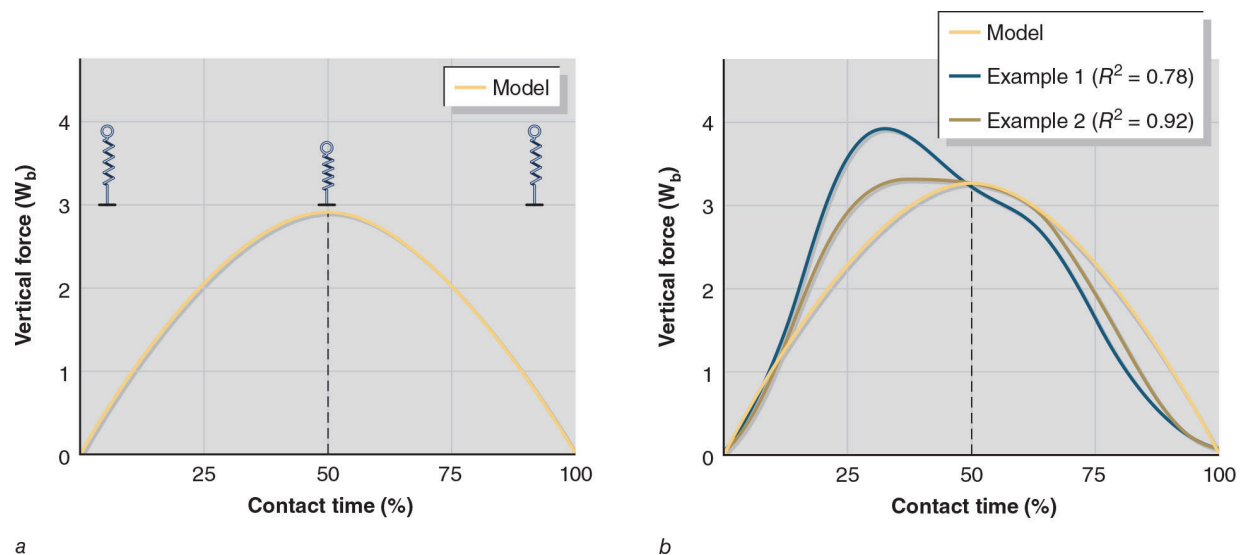


FIGURE 20.3 A simple spring–mass model relative to the ground reaction force during the stance phase of a sprint. During the stance phase, the model demonstrates how the leg (represented as a spring) is uncompressed at initial contact and then is compressed (represented by the change in length of the spring) during midstance or as vertical ground reaction force increases.

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Within an upright sprint, compression of the spring begins at foot strike, resulting in horizontal braking forces. This sudden and brief deceleration assists in propelling the swing leg forward in preparation for the subsequent step. As the center of mass moves ahead of the foot, the sprinter is in midstance. Within the SMM, the spring is compressed to the lowest point, which coincides with a lowered center of mass at midstance. Finally, the model describes

the push-off segment of the stance phase as the return of energy through the extension of the coiled spring. This resultant energy and return of force projects the sprinter forward.

While the SMM provides a conceptual framework for highlighting the actions involved in upright high-speed running, recent investigations suggest that the model's ability to describe the stance phase of elite sprinters is limited. As illustrated in [figure 20.2](#) on [page 640](#), during the maximal-velocity phase, elite sprinters tend to deviate from a classic SMM by producing much of their vertical force in the first half of a ground contact. In contrast, most non-elite sprinters, such as those involved in many team or field-based sports, display stance phases that are described by the SMM (24), where the vertical force curve is more symmetrical, as shown in [figure 20.3](#). As such, the SMM should be used as a means to describe the relationship between the SSC, muscle stiffness, and sprinting. In fact, as stride frequency increases at a given running speed, one of the most important features of the leg spring is an increase in muscle stiffness (39).

Because sprinting requires an athlete to move at high speeds, strength and conditioning professionals should emphasize the prescription of exercises that have been shown to increase neural drive while overloading musculature of the hip, knee, and ankle regions involved in the SSC.

Additional Neurophysiological Considerations for COD and Agility Development

In addition to the neurophysiological aspects of speed performance previously discussed, other factors must be considered regarding COD and agility performance. During the plant phase, depicted in [figure 20.4](#), the length of ground contact time of either an agility

(0.23-0.25 s) (7) or a COD movement (~ 0.15 -0.6 s dependent on angle) (8, 49, 64, 36) exceeds the typical ground contact time of both the acceleration phase of sprinting (0.17-0.2 s) (4) and the maximal-velocity phase of sprinting (0.09-0.11 s) (104, 105). In the context of COD, as the intended angle of direction change (i.e., deflection of center of mass) increases, the ground contact time of the main execution plant foot typically increases, attributed to the greater braking force, negative work (i.e., knee flexion), and deceleration demands associated when having to reduce momentum with sharper directional changes (36). As a result, this prolongs the contact time to increase braking impulse to reduce momentum along the original direction followed by propulsive impulse to enable acceleration toward the new intended direction of travel. In view of this, COD actions <45 degrees generally can be considered fast SSC activities, whereas direction changes >45 degrees generally can be considered slower SSC activities, though this depends on entry velocity and an athlete's physical capacity (36).



FIGURE 20.4 Plant phase, termed *stance phase* in the context of sprinting, of a COD movement. This is the point in a COD movement that represents the transition between the deceleration step and the acceleration step and is the main step in achieving deflection of the center of mass. Body positioning and the ability to maintain strong trunk positions during the deceleration of momentum and reorientation of the body to run in a new direction are critical for performance.

Since effective braking is an important part of agility performance (83, 84), neuromuscular development with respect to high-velocity and high-force eccentric muscle actions should be considered, for two reasons. First, the adaptations or motor unit recruitment pathways called on during an eccentric muscle action are different than those called on during concentric contractions (38). Secondly, the adaptations to eccentric training appear to be specific to the velocity of eccentric loading (74). In addition, training athletes for effective agility performance requires knowledge of perceptual-cognitive demands over and above the neurophysiological requirements for changing direction. Not only are the perceptual-cognitive demands on athletes related to their abilities in the areas of visual search scanning, anticipation, decision making, and reaction

time (87) but also the tactical situation (offensive vs. defensive) changes the brain processing strategy required (93, 96).

The training required to improve agility becomes clearer as one begins to understand the various demands required for COD and agility with respect to neurophysiological factors, including the SSC, eccentric muscle action neuromuscular training, and implications of the length of time or the demand of the COD maneuver on the SMM. Further, neurophysiological requirements of agility performance extend beyond physical requirements to perceptual-cognitive requirements that are specific to the tactical situation.

Running Speed

Sprinting is a series of coupled flight and support phases, known as **strides** (e.g., right foot contact to next right foot contact, essentially comprised of two steps—right to left, left to right), orchestrated in an attempt to displace the athlete's body down the track at maximal acceleration or velocity (or both), usually over brief distances and durations. Sprinting has been described as rapid, unpaced, maximal-effort running of 15 seconds or less (78). However, the classic definition of sprint speed concerns the relationship between **step length** (distance between the toe and heel of two consecutive footsteps) and **step frequency** (or rate; number of steps per second) (63). For clarity, one stride is equal to two steps (right to left, then left to right).

Based on this understanding, sprint speed can be increased by an increase in step length or an increase in step frequency ([figure 20.5](#)). Although these changes in performance variables are logical, the underlying component to maximizing step length and step frequency is related to rapid force production.

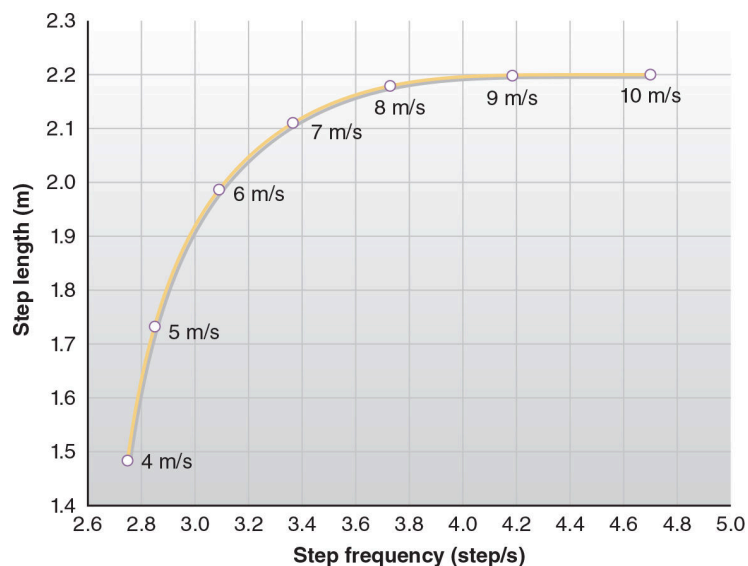


FIGURE 20.5 Step length–frequency interaction as a function of running velocity.

Adapted from Dillman (1975).

The differences between elite and novice sprinters can be traced to a single component. Much of the current literature on sprinting (17, 105, 106) suggests that the amount of vertical force applied to the ground during the stance phase at top speed may be the most critical component to improving top speed. In addition, these greater forces must be applied to the ground in the shortest period possible.

Application of force is needed to displace a mass. In sprinting, step length represents the displacement of mass. Elite male sprinters achieve a step length of 2.70 m (8.9 ft), whereas novice sprinters display a step length of 2.56 m (8.4 ft) at maximum velocity ([figure 20.6a](#)) (62).

Since contact with the ground is needed to continue force production and subsequent alterations in velocity, increasing the step frequency theoretically would maximize the time available to produce force. Elite male sprinters demonstrate step frequency near 4.63 steps per second compared to novice sprinters, who produce a lesser step frequency of 4.43 steps per second ([figure 20.6b](#)) (62). In other words, elite sprinters need less ground contact time with each step to exert the effort needed to displace their mass due to their ability to create greater impulses. Thus, these faster sprinters also spend more time in the air. Interestingly, elite sprinters display times to reposition the swing leg that are similar to those of their slower counterparts (62, 106).

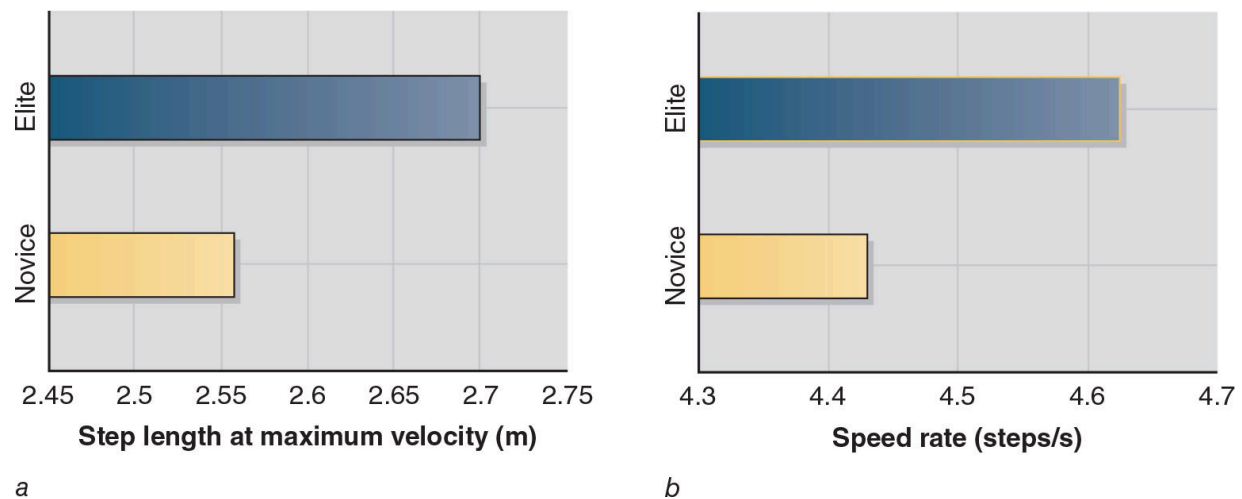


FIGURE 20.6 Step length (*a*) and step rate (*b*) in sprinters with varying qualifications.

Data from Mann (2011).

Although the time required for staging the subsequent ground contact is similar between elite and novice sprinters, elite sprinters are able to propel themselves farther down the track due to properly directed vertical forces. Mann (62) suggests that these vertical forces are better directed toward the track due to an optimized knee height at maximal flexion of the recovering leg. This higher knee position provides a greater time period for force production and subsequent ground clearance (i.e., range of motion force movement principle). This technical advantage may be further evidence for why elite sprinters tend to produce most of their force within the first half of the stance phase.

Additionally, faster sprinters are able to achieve higher velocities through the continuous application of high forces in a short stance phase, which results in longer steps or steps occurring at a higher rate. Within the elite setting (e.g., 100 m world championship finalists), this results in male sprinters achieving velocities between 11.0 meters per second and 12.5 meters per second (24.6-28.0 mph) (46). Conversely, team sport athletes generally achieve maximum velocities of approximately 8.0 to 9.5 meters per second (17.9-21.3 mph) (25). While force production is arguably the limiting

factor, technical efficiency and properly designed training also limit sprint speed.

Sprint speed is determined by an athlete's step length and step frequency; more successful sprinters tend to have longer step lengths as a result of properly directed forces into the ground while also demonstrating a more frequent step rate. These findings suggest that RFD and proper biomechanics are two of the primary limiting factors influencing sprint performance.

Sprinting Technique Guidelines

Linear sprinting is composed of a series of subtasks that can be divided into the start, acceleration ([figure 20.7](#)), and top speed ([figure 20.8](#)). While these phases of a sprint are technically distinct, they all require the athlete to volitionally move the lower limbs at maximal speeds through a series of stance and flight phases. The stance phase can be broken down further into a braking period followed by a propulsive period. In contrast, the flight phase consists of the **recovery** and **ground preparation** (i.e., late swing) segments of the swing leg.



FIGURE 20.7 Sprinting technique during the initial acceleration (i.e., the start) and the acceleration, touchdown, midstance, and takeoff of subsequent steps.



FIGURE 20.8 Sprinting technique at maximum velocity includes the toe-off, maximal vertical projection (MVP), touchdown/midstance, and subsequent toe-off phases.

[Figure 20.9](#) is a sprint technique checklist that accompanies [figures 20.7](#) and [20.8](#). These recommendations are based on linear sprinting and may be useful for both instruction and evaluation of

the movements. [Figure 20.10](#) on [page 649](#) describes the fundamental movements occurring in maximum-velocity sprinting.

Technical Errors and Coaching

[Table 20.1](#) highlights many common sprinting errors as well as possible causes and coaching corrections. Quite often, these errors are the result of the misapplication of forces due to improper coaching cues, insufficient mobility, or disruption to an athlete's normal gait caused by external interference. For instance, an athlete may demonstrate overstriding due to a coaching drill (such as reaching to successively longer marks along the track to gain speed) or the belief that acutely increasing step length will allow one to catch up to a competitor in a race. Regardless of the reason, the aim of strength and conditioning professionals should be to promote an athlete's speed through the proper transmission of forces into the ground that will optimize the athlete's gait cycle.

Figure 20.9 Sprinting technique checklist: start, acceleration, and maximum velocity.

Start

- The athlete should attempt to distribute a balanced body weight through a set position (blocks, three- or four-point staggered start) that uses segment angles critical to produce forces necessary to deliver explosive forces from the start.
 - Front lower leg angle should be approximately 90 degrees.
 - Rear lower leg angle should be approximately 120 to 135 degrees.
- Aggressive extension of the hip, knee, and ankle of the front leg against the block or ground should be observed.
 - After initially extending at the hip and knee, the rear leg should then be rapidly flexed at the hip (to ~90 degrees), knee (to ~90 degrees), and ankle (to neutral position) in preparation for foot contact of the first step.
 - As the rear leg is driven forward, the opposite arm should also drive forward, mirroring the range of motion and timing of the rear leg. The range of motion at the shoulder of each arm is large.

- The head should be neutral, looking a few yards or meters ahead along the track.

Acceleration

Posture

- Forward inclination helps maximize propulsive force contribution and depends on the absolute magnitude of force available. The forward lean gradually reduces with each step, assuming an upright position by maximum velocity.
- The pelvis position should be neutral relative to torso.
- The center of mass should be ahead of the foot during early acceleration (first few steps), which gradually progresses to the center of mass being slightly behind the foot as the athlete progresses to maximum velocity.
- The center of mass and hip position begin low and rise higher with each step from early to late acceleration to maximum velocity.

Arms

- The ranges of motion in the arms are wide to match the ranges of motion in the lower body on the contralateral side (e.g., as the knee travels forward, the opposite arm travels forward).
- The arms are driven aggressively backward.

Legs

- During acceleration each step gets progressively longer, with ground contact times getting shorter and flight times longer as the athlete advances to maximum velocity.
- Rapid, large split of the thighs through the stance phase enhances leg extension force while preparing the swing limb for rapid acceleration back down to the ground, enabling high ground reaction forces (GRF) early during ground contact.
- Low heel recovery enables shorter steps and more cycle time dedicated to force production. Heel recovery gains more height as the athlete progresses to maximum velocity.
- The path of the heel and rest of the foot tends to be piston like (backward and downward) before progressing to a more cyclic action by maximum velocity.
- Forward-oriented shin angles gradually become more vertical as the athlete progresses to maximum velocity.
- There should be rapid deceleration of extending the thigh (back-side catch) during the late stance to avoid prolonging ground contact and then enable rapid subsequent recovery of the leg after takeoff.

- The thigh forcefully retracts during the late swing to enable foot contact behind the center of mass, enhancing production and direction (more horizontal) of GRFs early in ground contact.
- The foot and ankle maintain optimal stiffness. A stable foot position during the early stance helps provide the foundation for the hip extensors to contribute more effectively to forward propulsion while enabling storage and release of elastic energy at the ankle.

Maximum Velocity

Posture

- Slight forward lean
- Head in neutral position looking toward intended direction of travel
- High neutral pelvis position

Arms

- Arm action mirrors the contralateral limb (i.e., when the touchdown leg is in its longest position the opposite arm will be in its most open position to balance the body).
- The hand is driven hard down below the waist on the downstroke and returns to the height of the athlete's chin with a more acute angle at the elbow (approximately 90 degrees on the return).
- Arms should not cross the midline of the body.

Legs

- With a slight flex of the knee at takeoff, the hip is in a hyperextended position and the ankle is plantar flexed.
- After toe-off, the knee should flex and the ankle dorsiflex (passive folding) as the thigh flexes forward at the hip to a high knee position. A high heel recovery whereby the foot steps over the opposite knee should be observed.
- With the knee at its highest point (front-side catch), the tibia should be in a high forward position, providing space for rearward acceleration of the thigh to help generate larger initial GRFs during support.
- The foot should travel down and back (negative foot speed before touchdown).
- The foot should touch down close to underneath the center of mass (will be slightly in front). At touchdown the opposite knee should be beside or slightly in front of the knee of the touchdown leg.
- During support, knee and ankle flexion should be minimal.
- In the late stance (back-side catch), early hip flexor activation helps avoid prolonged ground contact and decelerates the extending limb, preventing lower leg recovery behind the body after toe-off (too much back-side

mechanics), providing more optimal leg recovery (better timing and improved front side position).

- Efficient rapid switching of limb positions requires efficient timing, enabled by rapid switching off of key muscles; thus, relaxation and rhythm are essential to enable optimal timing and coordination and reduce energy demands.

Adapted from Mann (2011).

Training Goals

The overarching goal of sprinting is to achieve optimal step length and step frequency through the correct application of force into the ground. The transmission of high forces must occur within a short stance phase, also known as *ground contact time*. During the acceleration phase, propulsive forces assist in elevating the hips above the ground to a point at which speed is no longer increasing at large rates. This portion of the sprint, known as *maximum velocity*, uses the SSC (through stiffness regulation) to propel the athlete's center of mass down the track horizontally.

Figure 20.10 Fundamental movements occurring in maximum-velocity sprinting.

Early Flight

- *Eccentric hip flexion*. Decelerates backward rotation of the thigh
- *Eccentric knee extension*. Decelerates backward rotation of the leg and foot
- *Eccentric ankle dorsiflexion*. Decelerates ankle plantar flexion

Midflight

- *Concentric hip flexion*. Accelerates thigh forward
- *Eccentric knee extension* followed by *eccentric knee flexion*
- *Concentric ankle dorsiflexion*. Assists in reducing the limbs' moment of inertia

Late Flight

- *Concentric hip extension*. Rotates the thigh backward in preparation for foot contact

- *Eccentric knee flexion.* Accelerates the leg backward, limiting knee extension; stops before foot strike (aided by concentric knee flexion to minimize braking at touchdown)
- *Ankle in neutral position (avoid plantar flexed position).* Preparation for ground contact to reduce Achilles' tendon complex slack at touchdown to optimize storage of elastic energy during early support

Early Support

- *Continued concentric hip extension.* Minimizes the braking effect of the foot strike
- *Brief concentric knee flexion followed by eccentric knee extension.* Absorbs the landing and facilitates power transfer from the hip to the ankle during propulsion
- *Eccentric plantar flexion.* Stores potential elastic energy for subsequent release to facilitate greater propulsion during late support

Late Support

- *Eccentric hip flexion.* Decelerates backward thigh rotation; rotates the trunk in preparation for forward takeoff
- *Concentric knee extension.* Propels the center of gravity forward
- *Concentric plantar flexion.* Aids in propulsion

Based on Bezodis et al. (2008); Putnam and Kozey (1989); Wood (1987).

TABLE 20.1 Common Sprinting Technique Errors, Causes, and Corrections

Error	Cause	Coaching recommendations
START AND ACCELERATION		
Hips are too high in start of crouch position.	Misunderstanding of setup	Instruct athlete to space feet by 1.5 to 2 foot lengths, then lower into starting position by dropping the shin of the back leg to be parallel with sprint surface.
The athlete is stepping out laterally during the initial drive phase.	Improper distribution of forces	Instruct the athlete to push or drive through the ground to initiate the sprint.

*See sprint drills 1 to 6 on [pages 675-683](#).

Error	Cause	Coaching recommendations
The athlete's arm movement is abnormally short and tight.	Misunderstanding of natural arm swing	Instruct the athlete to either (a) drive the elbow down and back or (b) pull the hands down and back to simulate pulling on a rope. In addition, cue the athlete to allow the hands to fully break the waist while also allowing the arms to recover at midline of the body (invisible line from nose through navel).
The athlete is holding unnecessary tension in dorsal muscles and hyperextending the neck.	Misunderstanding of movement	Instruct the athlete to keep the head in line with the spine; the torso and head should rise at the same rate during the acceleration and transition phases of the sprint.
The athlete jumps the first step or steps over the knee of the stance leg.	Push-off angle too high; upward thrust too steep	Instruct the athlete to initiate movement by driving through the ground and allow the swing leg to horizontally cut the stance leg shin (i.e., ankle crossover), rather than stepping over the stance leg (i.e., knee crossover). In addition, cue the athlete to keep the foot of the swing leg close to the ground to set up a proper acceleration phase.
The athlete gets into an upright posture prematurely.	Inadequate push-off force; improper carriage of head	Instruct the athlete to continue pushing into the ground while maintaining a natural trunk lean. Also cue the athlete to keep the head in line with the spine; lifting the head may result in a sudden lifting of the torso, ultimately minimizing acceleration patterns.
MAXIMUM VELOCITY		
The athlete is superficially attempting to maintain an acceleration phase when the shins are clearly vertical.	Improper understanding of movement patterns	Instruct the athlete that as the shins and hips come up to vertical, so should the torso and head. Encourage the athlete to feel for the rise in the hips so that the joints (shoulders to hips to ankles) stay stacked or in line. This position allows for the proper transmission of forces into the running surface.

*See sprint drills 1 to 6 on [pages 675-683](#).

Error	Cause	Coaching recommendations
The athlete is not displaying optimal front-side mechanics with regard to the height of the swing leg knee.	Inadequate force production and hip mobility	Recall that the swing leg's knee height (traditionally called front-side mechanics) is influenced by GRFs and hip mobility. Improperly cueing an athlete to lift the knees may result in further improper transmission of forces and ultimately change the musculature naturally used during the sprint event.
The athlete is overstriding.	Misunderstanding of force application	Success in sprint events results from the ability to produce high vertical forces in a short amount of time. An overstriding athlete is attempting to increase speed via larger ground contact times, which ultimately dampens the effects of the stretch-shortening cycle and lowers step frequency. Instruct athletes to run in their lane and maintain their natural gait cycle.
The athlete is displaying chronic hamstring injury or pain.	Insufficient mobility; improper positioning of pelvis	A high likelihood of hamstring injury occurs during the sprint event's swing phase due to eccentric (lengthening) forces or overstriding. An athlete is further compromised when displaying anterior pelvic tilt during the sprint event. Reduce anterior pelvic tilt with appropriate development of back-side and front-side mechanics through appropriate sprint drills.*
The athlete is attempting to cycle the leg action, resulting in an increased time to complete the swing phase. This is made apparent by the open gap between the knees during the stance phase.	Improper force application	Instruct the athlete to drive the foot down and back into the track, not paw. Due to the seemingly horizontal movement of the foot during a sprint, coaches often miscue athletes to paw the foot into the track. Pawing the foot horizontally against the track prevents the athlete from using vertical forces, overloading the stretch-shortening cycle to move down the track. Integration of appropriate sprint drills* should help develop the correct leg action.
The athlete is displaying erroneous	Improper understanding of	While traditionally coaches would claim that erratic arm movement is a symptom of

*See sprint drills 1 to 6 on [pages 675-683](#).

Error	Cause	Coaching recommendations
arm movement in the transverse plane.	movement pattern	fatigue, much literature suggests that speed is limited by mechanical force application, not metabolic efficiency. Therefore, athletes should emphasize driving the arms down and back while maintaining an upright torso. In addition, strength and conditioning professionals should recommend to the athlete that the arm swing recovers near the midline of the body to take advantage of the glenohumeral joint's natural range of motion.

*See sprint drills 1 to 6 on [pages 675-683](#).

The following fundamental training objectives regarding performance enhancement and injury prevention should be considered during the development of speed in practical settings:

- *Emphasize brief ground support times as a means of achieving rapid step rate.* As previously mentioned, this requires high levels of ballistic strength. This quality is developed systematically through consistent exposure to speed training as well as properly designed resistance training programs.
- *Emphasize the further development of the SSC as a means to increase the amplitude of impulse for each step of the sprint.* Specifically, high achievers at top-speed sprinting produce high forces in a shorter stance phase using the SSC. The complete weightlifting movements and their derivatives are key exercises for overloading the SSC with forces greater than those produced during an open sprint, in addition to plyometric training, which can induce high-impact stretch loads to enhance SSC function and reactive strength.

Because sprinting requires an athlete to move at high speeds, strength and conditioning professionals should emphasize the prescription of exercises that span the

load–velocity curve while overloading musculature of the hip, knee, and ankle regions involved in the SSC.

Methods of Developing Speed

From a practical standpoint, the demonstration of maximal speed is the result of well-organized programming that organically develops and matures the required skill sets within the athlete. This organic approach to sprint training results from a schema of development that is based on the emphasis and de-emphasis of certain qualities, such as acceleration and maximum velocity, within each training phase. Furthermore, the training plan should coordinate these singular qualities through phasic progression. A well-constructed training plan highlights specific components that will assist in fully maximizing an athlete's movement potential.

Sprinting

Although a variety of training stimuli are important for the optimization of athletic performance, it can be argued that no exercise improves running velocity more than maximum-velocity sprinting (i.e., primary methods). An athlete's sprint prowess depends on the generation of high forces in short time periods (62, 105). Neurological adaptations resulting from long-term training plans that emphasize maximal strength and movement velocity improve both RFD and impulse generation (1, 100). In the strength and conditioning setting, weightlifting movements and jump training are prescribed to develop RFD and impulse at varying loads, because these movements use the SSC (i.e., tertiary methods). Similarly, upright sprinting uses the SSC, defining it as a plyometric movement. Chronic exposure to movements eliciting the SSC can increase muscle stiffness, which is a potential physiological advantage for sprint ability (39).

In addition, sprinting requires near-maximum to maximum muscle activation, which depends on high central nervous system activity. Therefore, chronic exposure to sprinting may lead to improvements in musculoskeletal control via the central nervous system. This would result in a cyclical dose-response relationship, because

neurophysiological adaptations from previous practices would bolster subsequent training. In an effort to either enhance the force required or target potential neuromuscular adaptations for sprinting improvement, resisted and assisted sprint training techniques are often employed (i.e., secondary methods). The proposed benefits, potential drawbacks, and coaching considerations are outlined in [table 20.2](#).

Strength

As noted throughout the chapter, sprint speed is underpinned by an athlete's ability to produce large forces within brief time periods. These forces must be large enough to support the body weight in the presence of gravity and displace the body through an increase in velocity (62). Thus, many strength and conditioning professionals are aware of the importance of resistance training in the development of a sprint-based athlete. Central to the discussion of resistance training is how best to transfer newly developed strength qualities from the weight room to the track or field (109). The transfer of strength improvements to sprinting may require an emphasis on the specificity of training (i.e., **dynamic correspondence** and **coordinative overload**). This transfer of training effect deals with the degree of performance adaptation and may result from the similarities between the movement patterns, peak force, RFD, acceleration, and velocity patterns of an exercise and the sporting environment (98).

While training for maximal strength may be beneficial, training should emphasize agendas that merge maximal strength and speed–strength qualities instead (44). The selection of exercises and movements that provide opportunities to display forces and velocities like those found in sprinting may have the most benefit with regard to enhancing rate coding and firing frequency, alongside alterations to type II muscle fibers such as cross-sectional area and fascicle length (40, 43). For instance, evidence suggests that weightlifting movements and their derivatives such as the clean,

snatch, and mid-thigh pulls may enhance sprint performance through physiological adaptations such as muscular stiffness, enhanced RFD, and coactivation of the musculature surrounding the hips and knees (6, 26, 32).

Mobility

While not a kinetic variable, soft tissue manipulation is a commonly used practice in the development of speed athletes. Strength and conditioning professionals and athletic trainers rely on tools such as stretching, chiropractic care, massage, and myofascial release to achieve optimal mobility within the dynamic state. Mobility is the freedom of an athlete’s limb to move through a desired range of motion, whereas flexibility is a joint’s total range of motion. With an understanding that positional characteristics are among several limiting factors in performance, strength and conditioning professionals should ensure that proper postural integrity is in place before the onset of practice or competition.

TABLE 20.2 Assistance and Resistance Training for Speed Development

Type of training	Example exercise modalities	Potential benefits	Potential disadvantages	Practical use suggestions
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Note: SSC = stretch-shortening cycle.

Type of training	Example exercise modalities	Potential benefits	Potential disadvantages	Practical use suggestions
Assistance training (e.g., overspeed training)	Inclusive of modalities such as rope towing, bungee cord pulls, or downhill running with the intention of allowing the athlete to run at supramaximal speeds	- Theoretically, these training tools are used to potentially expose the athlete to higher velocities than one can achieve in unassisted sprinting by inducing an increased step or stride rate and exposing the athlete to higher hip angular velocities (46). - This increased step rate is hypothesized to have potential for neuromuscular adaptations to increase maximal sprint velocity. Increases in overspeed should be limited to $\leq 10\%$.	- Assisted sprinting may prematurely rush an athlete's stance phase, thus removing time needed to exert optimal forces, and has been shown to result in decreased muscle activation and propulsive force production in comparison to maximal running (66). Therefore, what can be described as chop steps may be observed. Undesirable techniques such as leaning back and overstriding may be present in a protective manner to brake, which is suboptimal. - Exercises that rely on the towing of athletes may increase braking forces in comparison to maximal running (66) as athletes attempt to	- Top-speed sprinting should be developed through more natural and safer training tools that consider an athlete's training status and support optimal biomechanics. - Strength and conditioning professionals should carefully assess the implications of assistance modalities before implementation and these modalities may be best used when working with advanced or elite athletes.

Note: SSC = stretch-shortening cycle.

Type of training	Example exercise modalities	Potential benefits	Potential disadvantages	Practical use suggestions
			tolerate velocities they cannot naturally create or accommodate because of biomechanical efficiency, training state, or both. - Downhill sprinting may expose the athlete to unnecessary eccentric forces through a modified stance leg that results from having to find the ground. This may also affect the SSC, which is optimized through the foot placement occurring just in front of the center of mass on ground contact in flat-surface upright sprinting (66). - To date, very little research supports the implementation of overspeed training, because sprint success results from an	

Note: SSC = stretch-shortening cycle.

Type of training	Example exercise modalities	Potential benefits	Potential disadvantages	Practical use suggestions
			athlete's ability to produce large forces in a short ground contact time.	
Resistance sprinting	Inclusive of training modalities such as sled towing, sled pushing, wind resistance, incline sprinting, and sled pushing with the intention of improving an athlete's accelerative abilities	• Accentuates the biomechanics of the acceleration phase by placing the athlete in a position that promotes lowered torso (i.e., forward trunk lean), hips, and shin angles. • Resisted sprinting may optimize an athlete's ability to cover short distances quickly by overloading the acceleration phase, resulting in ground contact times that produce higher propulsive forces. • Improving acceleration, which is a change in	• Sprint efforts against a load that is too great may lead to longer ground contact times in addition to shorter step lengths that are not task specific (59). • Slopes of incline that are too great may modify proper sprinting biomechanics, resulting in a rehearsal of improper technique and thus limiting transfer of training effect. • Sled pushing may alter the natural sprinting gait cycle due to the removal of arm action that synchronizes and counterbalances the leg action occurring at the hip.	• Strength and conditioning professionals and athletes attempting to improve the acceleration phase of a sprint may adopt resistance training modalities that maintain normal biomechanics involved in unresisted sprints. • Loads should be chosen based on the sporting context while also considering the athlete's physical status. For instance, track and field sprinters may use loads that do not decrease running velocity by more than 10% to 12% (2, 49, 52, 61). • In contrast, field sport athletes

Note: SSC = stretch-shortening cycle.

Type of training	Example exercise modalities	Potential benefits	Potential disadvantages	Practical use suggestions
		velocity, may lead to an increase in top speed through enhanced RFD. • Resisted sprints and loads, and velocity decrements, can be manipulated to target technical, speed–strength, power, and strength–speed qualities (19, 20). • It may offer specificity to sporting scenarios when athletes need to push (move) external mass such as in bobsled, rugby, and American football.		who overcome external resistance while blocking, tackling, and scrimmaging can use loads 20% to 30% of body weight to improve the initial 5-10 m of movement (28, 52). Frictional properties will influence the amount of resistance experienced, thus prescribing sled pushes and pulls based on velocity decrements is suggested (19, 20).

Note: SSC = stretch-shortening cycle.

Based on the previously described constructs of speed, the currently accepted model of sprint success is based on the athlete's ability to produce and overcome ground reaction forces in a short amount of time. Moreover, while these GRFs assist in producing optimal step length through forward propulsion, the sprinter's position in flight may be limited by insufficient mobility (i.e., reduced hip flexion restricts subsequent thigh and knee lift). Specifically, an

athlete may possess the physical characteristics necessary to yield high rates of force in a short amount of time, but compromises in a joint's freedom of movement will result in misplaced forces. The improper placement of forces due to erroneous ground contact will result in dampened sprint speed and an increased likelihood for injury.